SIGN LANGUAGE GLOVE USING ARDUINO

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BONAFIDE CERTIFICATE

Certified that this project report titled, "SIGN LANGUAGE GLOVES USING ARDUINO", for Microcontroller and Embedded System (2311EEC404R) is a bonafide work of **ABIRAMI.V(310624105004), ATSHAYA.K. N (310624105020), DHAYASSRI.R(310624105030), DIBIKA.A(310624105031)** who carried out the project work under my supervision. Certified further that to the best of my knowledge, the work reported here in does not form part of any other project report or dissertation.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Sign language is an important way for people with hearing and speech impairments to communicate, but it can create barriers with those who don't understand it. To help with this issue, we have created a Sign Language Glove using Arduino. This solution is innovative, affordable, and portable. It translates hand gestures into readable text or spoken words in real time, allowing sign language users and non-users to interact smoothly.

The glove uses multiple flex sensors placed along the fingers and hand. These sensors detect finger bending and hand orientation, turning physical gestures into electrical signals. An Arduino microcontroller processes these signals and links them to specific sign language gestures through embedded programming. Each gesture corresponds to a word, letter, or phrase stored in the system's database. When the Arduino recognizes a gesture, it sends the output to a display unit like an LCD or a mobile device via a Bluetooth module. A text-to-speech module can also be added to convert the recognized gesture into voice output, improving real-time verbal communication. The system is lightweight, energy-efficient, and user-friendly, making it practical for everyday use.

This project focuses on being affordable and simple to make assistive technology available to more people. Compared to systems that rely on image processing, the glove approach offers quicker response times, lower computing needs, and better accuracy in controlled settings. However, issues like a limited gesture vocabulary, calibration needs, and differences in hand sizes remain and can be resolved in future updates.

In conclusion, the Sign Language Glove using Arduino shows a useful way to apply embedded systems and sensor technology in assistive communication. It helps break down communication barriers and fosters inclusivity by allowing people with different abilities to engage more effectively in society. Future improvements may involve adding machine learning, expanding gesture recognition, and integrating cloud-based communication systems for better performance and scalability.

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CHAPTER 1

INTRODUCTION

1.1GENERAL INTRODUCTION

Communication is a fundamental human need, yet individuals with hearing and speech impairments often face significant challenges in interacting with others. Sign language is their primary mode of communication, but it is not widely understood by the general population. This creates a communication gap that can lead to social isolation and limited opportunities. To address this issue, technology-driven solutions are being developed to translate sign language into text or speech, making communication more accessible and inclusive.

The Sign Language Glove using Arduino is an innovative approach that utilizes embedded systems and sensor technology to recognize hand gestures and convert them into understandable output. By integrating flex sensors and motion sensors with an Arduino microcontroller, the glove captures finger movements and hand positions, processes the data, and displays the corresponding message. This system provides a portable, cost-effective, and efficient solution for real-time communication, helping bridge the gap between sign language users and non-users.

1.2 OBJECTIVES

The primary objectives of this project is:

1. To design and develop a smart glove capable of recognizing sign language gestures using sensors.
2. To convert hand gestures into readable text using an Arduino microcontroller.
3. To enable real-time communication between speech/hearing impaired individuals and others.

1.3 EXISTING SYSTEM ANALYSIS

Currently, communication for hearing and speech impaired individuals mainly relies on traditional sign language, which requires both the sender and receiver to have prior knowledge of the language. This creates a major limitation, as most people are not trained in sign language, leading to communication gaps in daily interactions. In many situations, interpreters are required, which may not always be available, making communication inconvenient and sometimes impossible.

Existing technological solutions include vision-based systems that use cameras and image processing techniques to recognize gestures. While these systems can be effective, they are often expensive, require high computational power, and are sensitive to lighting conditions and background noise. Some systems also suffer from delays in processing and lack portability. Compared to these methods, sensor-based glove systems offer a simpler, more cost-effective, and portable alternative, though they may have limitations in recognizing complex gestures and require proper calibration.

1.4 PROPOSED SYSTEM

The proposed system is a smart glove-based assistive device that converts hand gestures of sign language into text and speech output using sensors and a microcontroller. It helps bridge communication between hearing/speech-impaired people and normal users.

Central Controller :Arduino Uno

The entire system is centered around an Arduino Uno, which acts as the control unit of "brain" of the project. It's a microcontroller board that runs the custom software, processes all inputs, and makes decisions based on the programmed logic.

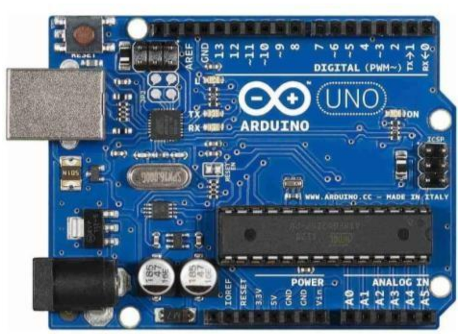


Fig1.1ArduinoUno

Flex sensor

A flex sensor is a variable resistor that changes its resistance when bent. In its normal (straight) position, it has low resistance, and as it bends, the resistance increases due to the stretching of its conductive layer. This change in resistance is converted into a corresponding voltage using a voltage divider circuit, which can be read by a microcontroller like Arduino through an analog input.

Fig1.2 Flex sensor

Resistor:

A resistor is a basic electronic component that opposes the flow of electric current in a circuit. Its main function is to control or limit the amount of current and to divide voltage as required. The resistance is measured in ohms (Ω). Resistors are commonly used to protect components from excessive current, set operating conditions, and form voltage divider circuits

Fig1.3 Resistor

LCD Display:

The LCD display in the sign language glove system is used to present the recognized gestures in the form of readable text. After the microcontroller processes the sensor data and identifies the corresponding gesture, the output is sent to a 16×2 LCD module, which displays the message instantly. This enables direct communication for the user, allowing others to easily understand the intended message without the need for additional devices.

Fig1.4 LCD Display

Breadboard:

The breadboard in the sign language glove system is used for building and testing the circuit without soldering. It allows easy connection of components like the microcontroller, sensors, LCD, and other modules using jumper wires. The breadboard helps in organizing the circuit, making modifications simple, and ensuring proper electrical connections during prototyping and testing

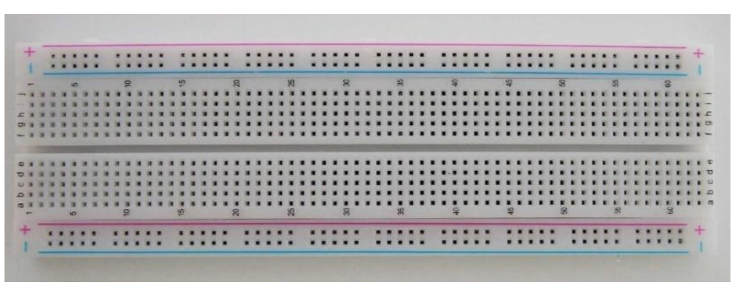


Fig1.5 Breadboard

Bluetooth:

The HC-05 Bluetooth Module is a wireless device used to connect microcontrollers like Arduino to smartphones via Bluetooth. It uses serial communication (UART), works in master/slave modes, and is commonly used for robotics and home automation.

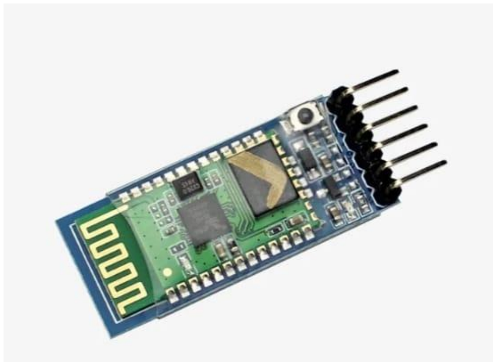


Fig1.6 Bluetooth

CHAPTER 2

LITERATURE SURVEY

- Khan and Chahar (2025) designed a real-time sign language to speech conversion glove using Arduino, flex sensors, and an accelerometer. The system translates hand gestures into audible speech via Bluetooth, enhancing accessibility and enabling seamless interaction for hearing-impaired users.
- Nwachukwu et al. (2024) implemented a smart glove system using flex sensors to detect hand gestures in sign language. Their work focuses on wearable sensor integration and gesture recognition, demonstrating an effective method for translating sign language into understandable outputs.
- Gudi et al. (2024) proposed a real-time sign language translation system using Arduino Nano, flex sensors, and a Bluetooth module. The system integrates deep learning models for gesture classification and uses text-to-speech technology to produce voice output, improving communication accuracy.
- Manoj et al. (2023) designed an Arduino-based hand gesture recognition device that translates sign language into text or speech. Their system uses flex sensors and microcontroller processing to interpret gestures, providing a low-cost communication solution.
- Gautam et al. (2023) presented a review on smart sign gloves using microcontrollers like Arduino and ESP32. Their study emphasizes the role of sensors and machine learning in improving gesture recognition and enabling real-time communication systems

2.1PROBLEM STATEMENT

- Deaf and mute individuals face difficulty in communicating with people who do not understand sign language.
- There is a lack of common communication medium between sign language users and normal speakers.
- Sign language is not widely understood, creating barriers in daily interactions like hospitals, schools, and public places.
- Existing solutions like interpreters are not always available and can be costly.
- Communication delays can lead to misunderstanding and frustration.
- Current systems are often complex, expensive, or not portable.
- There is a need for a real-time, affordable, and user-friendly system.
- Manual translation of gestures is time-consuming and inefficient.
- Limited technological solutions for gesture-to-speech conversion in everyday use.
- Need for a device that can accurately detect hand gestures and convert them into text or voice instantly.

CHAPTER 3

PROPOSED WORK

3.1 INTRODUCTION

The proposed system focuses on developing a sign language recognition glove using Arduino to assist communication between deaf or mute individuals and others. The system uses flex sensors attached to the fingers to detect hand gestures by measuring the bending of each finger. These sensor signals are processed by the Arduino, which compares them with predefined patterns to recognize specific gestures. Once a gesture is identified, it is converted into text displayed on an LCD or speech output through a speaker.

This project aims to provide a real-time, low-cost, and portable solution that simplifies communication and reduces dependence on sign language interpreters. By integrating simple sensors and microcontroller technology, the system offers an efficient and user-friendly way to translate hand gestures into understandable language.

3.2 METHODOLOGY

The methodology explains how the system is designed and works step-by-step from input to output.

1. System Design:
    - The system is divided into three main stages
    - Input Stage → Flex sensors detect finger movements
    - Processing Stage → Arduino processes the signals.
    - Output Stage → Displays text or produces speech
2. Sensor Placement and Setup
    - Flex sensors are attached to each finger of the glove.
    - Each sensor is connected to Arduino analog pins (A0–A4).
    - A voltage divider circuit is used to convert resistance into voltage.
    - Proper calibration is done for each finger

3. Data Acquisition
    - When the user performs a gesture:
    - Flex sensors bend
    - Resistance changes
    - Voltage output varies

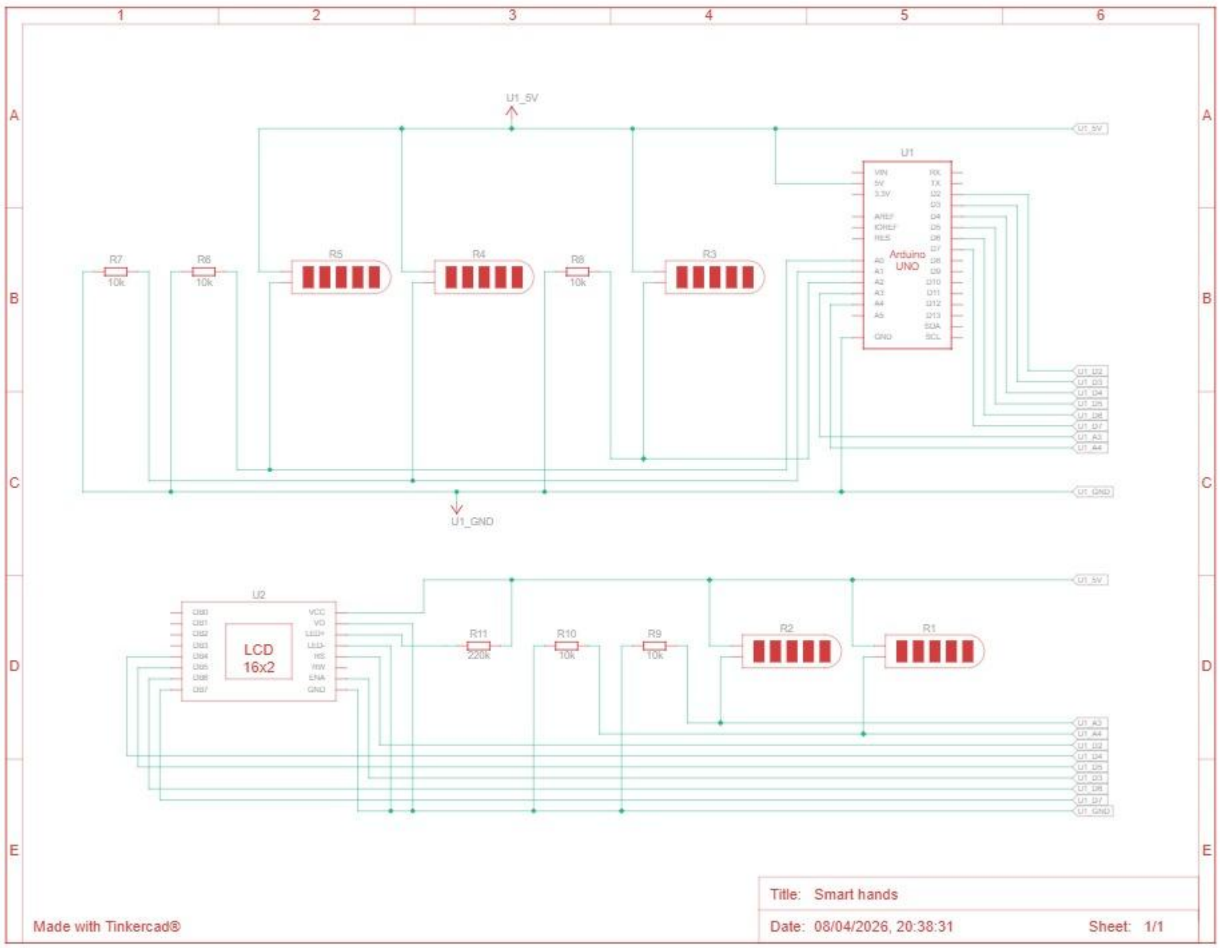
4. Signal Processing
    - The analog values are:
    - Filtered (to remove noise)
    - Compared with predefined threshold values

5.Gesture Recognition
    - A set of predefined gesture patterns is stored in the program
    - The system matches current sensor values with stored patterns
    - If a match is found:
        - Corresponding word/message is selected

6. Output Generation
    - Text Output
    - Displayed on LCD (e.g., “HELLO”)

7. Testing and Validation
    - Different gestures are tested
    - Accuracy and response time are observed
    - Adjustments are made for better performance

3.3 WIRING DIAGRAM OF THE INDUSTRY



CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

The system implementation of the sign language recognition glove using Arduino involves the practical realization of the designed model by integrating hardware and software components. In this stage, flex sensors are mounted on the glove to capture finger movements, and these sensors are connected to the Arduino through analog input pins using a voltage divider circuit. The Arduino is programmed to read the sensor values, process them, and compare them with predefined gesture patterns stored in the code.

Based on the recognized gesture, the system generates output in the form of text on an LCD display or audio through a speaker. Proper wiring, calibration of sensors, and testing are carried out to ensure accurate gesture detection and smooth operation. This implementation provides a real-time, efficient, and user-friendly solution for converting hand gestures into meaningful communication.

4.2 HARDWARE DESIGN

The hardware design of the proposed sign language glove system consists of integrating sensors, a microcontroller, and output devices to achieve real-time gesture recognition. Flex sensors are mounted on each finger of the glove to detect bending, and they are connected to the analog input pins of the Arduino using voltage divider circuits with resistors. The Arduino acts as the central processing unit, receiving analog signals from the sensors and converting them into digital values. A 16×2 LCD display is interfaced with the Arduino through digital pins to show the recognized gestures as text, while an optional buzzer or speaker is connected to provide audio output. All components are powered using the Arduino’s 5V supply, and proper grounding is maintained throughout the circuit. This hardware setup ensures accurate sensing, efficient processing, and reliable output generation in the system.

4.3 CORE COMPONENT: ARDUINO MICROCONTROLLER

The microcontroller is a compact integrated circuit that acts as the brain of the entire system, governing all its operations. It is chosen for its ease use,extensive community support ,and sufficient number of digital I/O pins to control all the components .and great hero this project is it's built-in ,which is a one volatile memory required to store slipcases he persistent lockout counter ,ensuring data is retained even when power is disconnected.

4.3.1 Functions of the Microcontroller

The. main functions of the Arduino Uno in this project are:

- To continuously read data from flex sensors and motion sensors to detect hand and finger movements.
- To process and convert analog sensor signals into digital values for further analysis.
- To compare the detected gesture patterns with predefined sign language data stored in memory.
- To generate corresponding output signals such as text or voice based on recognized gestures.
- To control output devices like LCD display, speaker, or mobile interface for communication..

CHAPTER 5

SIMULATION AND RESULTS

5.1 INTRODUCTION

This chapter introduces the simulation model developed for the Sign language gloves using Arduino using the Autodesk Tinkercad platform. It provides a dynamic environment to test how the system responds to different hand gestures and sensor inputs without using actual hardware. The simulation helps validate the accuracy of gesture recognition and the effectiveness of the programmed algorithms. It allows testing of all core features, including detection of different sign gestures, conversion into text or speech output, and system response time.

5.2 LOGICAL FLOW DIAGRAM OF THE SYSTEM

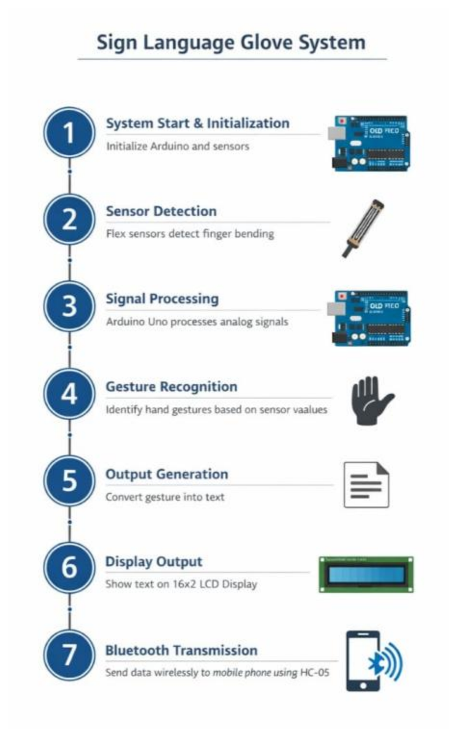
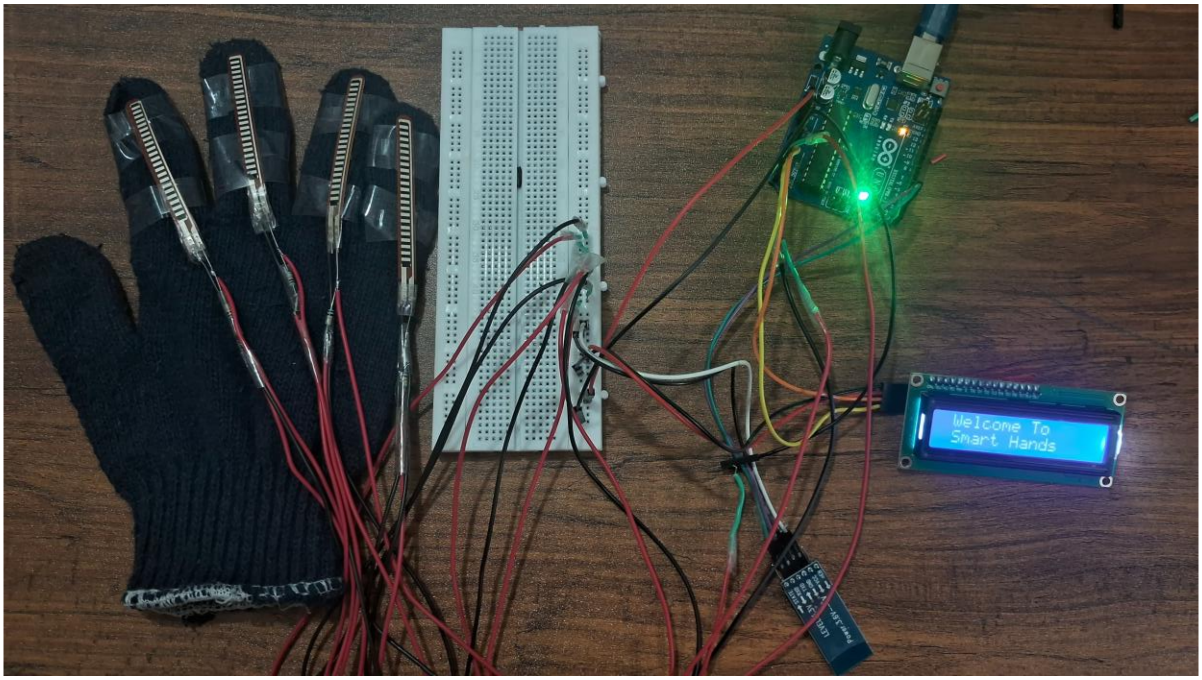


Figure 5.1FlowDiagram

5.3 EXPLANATION

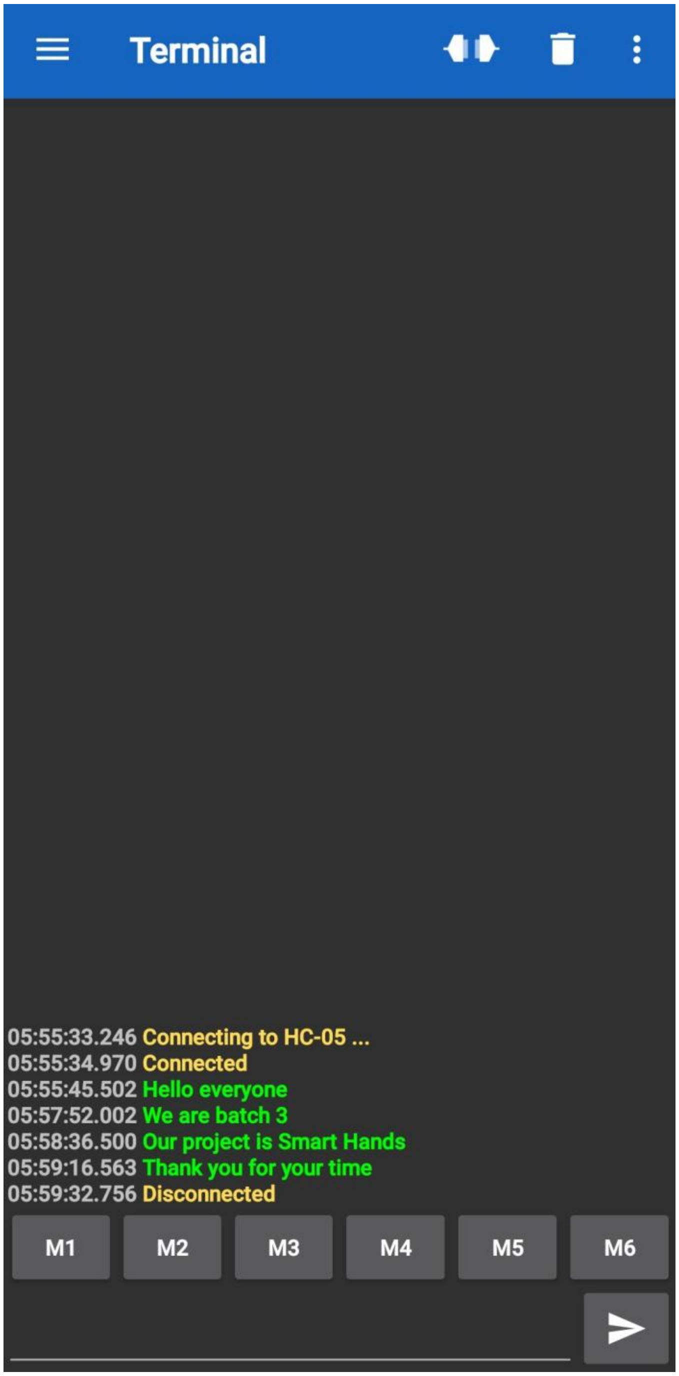
- System Start & Initialization: The system begins by initializing the Arduino microcontroller and all connected sensors. It ensures all components are ready to function properly.
- Sensor Detection: Flex sensors detect finger bending movements. Each finger movement produces a different electrical signal.
- Signal Processing: The Arduino reads the analog signals from sensors.
- These signals are converted into digital values for processing.
- Gesture Recognition: The system compares sensor values with predefined patterns. It identifies which hand gesture is being performed.
- Output Generation: Once a gesture is recognized, it is converted into text format.
- Display Output: The generated text is displayed on a 16×2 LCD display. This helps in direct communication.
- Bluetooth Transmission: The output is also sent wirelessly to a mobile phone using a Bluetooth module (HC-05).

5.4 HARDWARE SIMULATION AND RESULTS

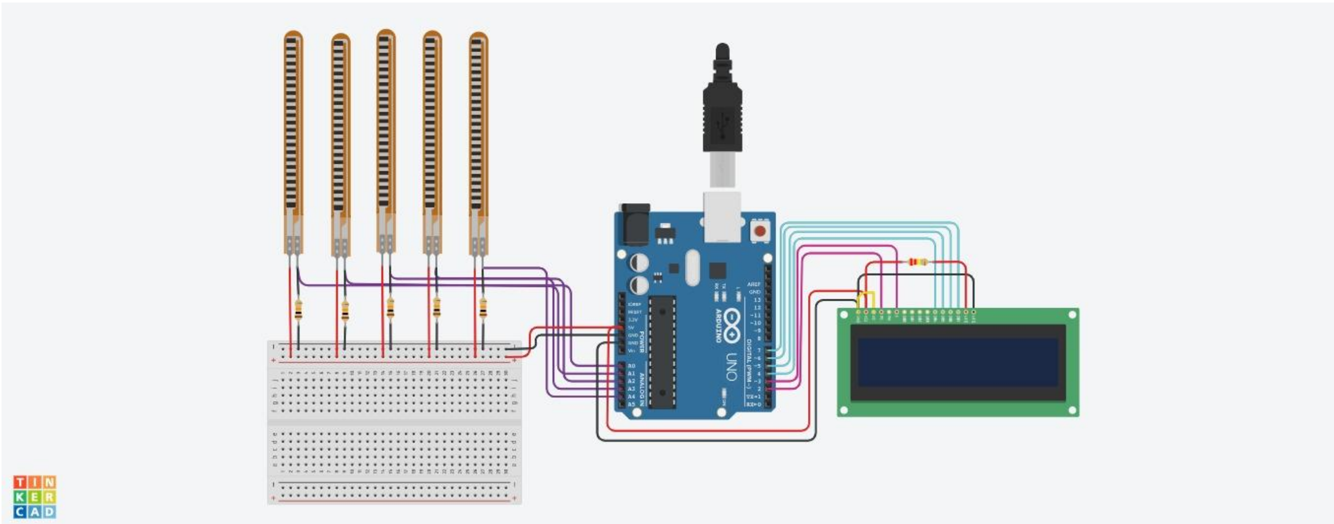


5.1.1 Hardware setup

- The project is based on a **flex sensor that detects bending or movement**.
- The flex sensor is connected to the analog pin of the **Arduino Uno**.
- When the sensor is bent, its **resistance changes**, producing different analog values.
- The **Arduino Uno** reads these analog values using its **analog input pins**.
- The processed values are displayed on a **16×2 LCD display**.
- The LCD shows the **real-time sensor readings continuously**.
- This project demonstrates **sensor interfacing, analog signal processing, and LCD display using Arduino**.
- Flex sensors are widely used in **robotics, gesture detection, medical devices, and wearable technology**.



5.1.2 Serial monitor output



5.5 TINKERCAD SIMULATION AND RESULTS

CHAPTER 6

CONCLUSION

The Sign Language Gloves using Arduino project successfully demonstrates an effective method of bridging the communication gap between speech-impaired individuals and others. By integrating flex sensors, an Arduino microcontroller, and a display/output system, hand gestures are accurately detected and converted into understandable text or speech.

This system proves to be cost-effective, portable, and user-friendly compared to existing solutions. It highlights the practical application of embedded systems and sensor technology in solving real-world problems. Although the current model supports a limited set of gestures, it can be further enhanced with advanced sensors and machine learning techniques for improved accuracy and a larger vocabulary. Overall, the project contributes to assistive technology by promoting inclusivity and enabling better communication for the deaf and mute community.

FUTURE SCOPE

- Increased Gesture Recognition: Expand the system to recognize a larger set of sign language gestures, including full sentences and complex expressions.
- Integration with Machine Learning: Use AI/ML algorithms to improve accuracy, adaptability, and real-time learning of user-specific gestures
- Multi-language Support: Enable translation into different languages to make the system globally useful.

CHAPTER 7

REFERENCES

1. Jhunjhunwala, Yash, et al. "Sign language to speech conversion using arduino." International Journal of Recent Innovation in Engineering and Research January-2017 (URIER) (2017): 37-44.
2. Küçükdermenci, Serhat. "Multifunctional Smart Glove: An Innovative Solution for Sign Language Interpretation and Wireless Wheelchair Control." 3rd International Conference on Frontiers in Academic Research. 2024.
3. Ahmed, Mohamed Aktham, et al. "A review on systems-based sensory gloves for sign language recognition state of the art between 2007 and 2017." Sensors 18.7 (2018): 2208.
4. Ambar, Radzi, et al. "Development of a wearable sensor glove for real-time sign language translation." Annals of Emerging Technologies in Computing (AETiC) 7.5 (2023): 25-38.
5. Thaim, Ahmad Imran Mohd, et al. "Smart Glove for Sign Language Translation." J. Adv. Res. Appl. Mech. 112.1 (2024): 80-87.
6. ZAKI, AHMAD KAMIL BIN MOHAMAD. "SMART GLOVE SIGN LANGUAGE." (2023).
7. Raj, Vinith. "Talking Gloves: Sign Language to Speech Conversion for Deaf and Mute Person." Journal of University of Shanghai for Science and Technology (2021).
8. Padhya, Dinanath, et al. "Design and Implementation of a Multi-Purpose Low-Cost Hall-Effect Sensor Glove for Sign Language Recognition." arXiv preprint arXiv:2601.11539 (2025).
9. Manideep, Jalamanchili Sai Gana, et al. "Developing a Sign Language Interpreter for Speech-Impaired People Using Arduino." IUP Journal of Telecommunications 15.1 (2023).
10. Bukhari, J., et al. "American sign language translation through sensory glove; signspeak." International Journal of u-and e-Service, Science and Technology 8.1 (2015): 131-142.
11. Amin, Muhammad Saad, Syed Tahir Hussain Rizvi, and Md Murad Hossain. "A

comparative review on applications of different sensors for sign language recognition." Journal of Imaging 8.4 (2022): 98.

12. Nayak, Akshay V., et al. "Design of smart glove for sign language interpretation using NLP and RNN." International Conference on Future technologies in Manufacturing, Automation, Design and Energy. Singapore: Springer Nature Singapore, 2020.
13. Bantval, Dhruv, and Charlene Xia. "Motion Sensor-based Gloves Converting ASL to Speech to Aid Hearing Impaired." Journal of the National Medical Association 116.4 (2024): 446-447.
14. Lakshmi, I. Rajya, et al. "Arduino and flex sensor based hand gesture to speech conversion." Dogo Rangsang Research Journal 13.4 (2023): 346-355.
15. Dadaram, Mali Pooja, et al. "Sign Language to Speech Conversion Gloves using Arduino and Flex Sensors." Sign 7.04 (2020).
16. Burhani, Ziad MHD Adib, and Judhi Prasetyo. "The sign language interpreting gloves." Journal of Applied Science and Advanced Engineering 1.1 (2023): 18-27.
17. Ghosh, Poulami, Ainesh Dutta, and Sumit Topno. "Sign language hand glove." American Journal of Electronics & Communication 3.1 (2022): 14-16.
18. Hasan, Walid KA, and Nadia Naji Gabeal. "Implementation smart gloves for deaf and dumb disabled." Int Sci Technol J 19 (2019).